Problem: Cube Edge Rotation Invariant

A cube has 1 red and 5 white faces. We place it on a table so that it rests on the red face.
An "edge move" is a 90° rotation of the cube around one of the four edges resting on the table, so that a new face will then be resting on the table.
Suppose somebody performs 12 edge moves, such that each one of the cube's 12 edges is used exactly once as the axis of rotation for an edge move.
Prove that after the final move, the cube will again rest on the red face.